

PART B — Chipzo Project Kits

Each kit is a complete bundle. One purchase, multiple buildable projects. No extra components needed. Designed for school, college, and hackathon students.

Kit 1 — Arduino Starter & Basics Kit		Perfect first kit · Ages 12+ · School to Diploma level
Components Included	Projects You Can Build (10)	
• Arduino Uno R3 (CH340) x1 • USB Type-A to Type-B cable x1 • Breadboard 830 point x1 • Jumper wires M-M 20cm (40 pcs) • Jumper wires M-F 20cm (40 pcs) • LEDs red, green, blue, yellow, white x10 each • Resistor pack 220Ω, 1kΩ, 10kΩ (20 each) • Capacitor 100nF ceramic x10 • Push-button tactile switch x5 • SPDT slide switch x2 • Active buzzer 5V x1 • 9V battery holder x1 • 16x2 LCD with I2C backpack x1 • DHT11 sensor x1 • HC-SR04 Ultrasonic sensor x1 • Relay module 1-channel x1 • Servo SG90 x1 • Potentiometer 10kΩ x2 • NPN transistor BC547 x5 • Diode 1N4007 x5	1. LED Blinking & Traffic Light Control LED sequences, learn digital I/O 2. Push-button Counter Count presses and show on LCD 3. Temperature Monitor Read DHT11, display on LCD 4. Ultrasonic Distance Meter Measure and display distance 5. Smart Light with Relay Auto-switch light on ambient brightness 6. Servo Angle Controller Control servo with potentiometer 7. Burglar Alarm Ultrasonic + buzzer + relay 8. Password Door Lock Button keypad + servo latch 9. Simple Weather Station Temp, humidity, heat index on LCD 10. Plant Watering Alarm DHT + buzzer for dry soil alert	

Kit 2 — IoT & Wi-Fi Smart Projects Kit		Cloud connect · Ages 15+ · Diploma / B.Tech semester projects
Components Included	Projects You Can Build (10)	

• ESP32 WROOM-32 Dev Board x1 • USB Type-A to USB-C cable x1 • Breadboard 830 point x1 • Jumper wires M-M and M-F (40 pcs each) • DHT22 Temp and Humidity sensor x1 • Soil Moisture Sensor module x1 • MQ-2 Gas Sensor x1 • PIR Motion Sensor x1 • Relay module 4-channel x1 • 0.96 inch OLED I2C display x1 • LED assortment + 220Ω resistors x10 • Active buzzer x1 • Push-button switches x3 • 18650 Li-Ion cell + holder x1 • TP4056 charging module USB-C x1 • Potentiometer 10kΩ x1	1. IoT Weather Station Send temp/humidity to Blynk/ThingSpeak cloud 2. Smart Home Automation Control 4 appliances via mobile app over Wi-Fi 3. Gas Leak Detector + Alert MQ-2 triggers notification via ESP32 4. Smart Security System PIR + OLED + relay + cloud push notification 5. Automatic Plant Irrigation Soil moisture auto-triggers water pump relay 6. Air Quality Monitor Dashboard MQ-135 + OLED + live web dashboard 7. Energy Monitor ACS712 + ESP32 logs power usage to cloud 8. Wi-Fi Attendance System RFID + ESP32 logs to Google Sheets 9. ESP32-CAM Smart Doorbell Motion photo capture + Telegram notification 10. Smart Parking Counter Ultrasonic sensors count slots on web UI
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Kit 3 — Robotics & Motor Control Kit <i>Build and drive · Ages 14+ · School robotics / hackathons</i>	
Components Included	Projects You Can Build (8)
• Arduino Uno R3 x1 • L298N Motor Driver Module x1 • BO Motor 150 RPM x2 • Robot Wheel 65mm x2 • Ball Caster Wheel 1 inch x1 • 2WD Acrylic Chassis base x1 • Breadboard 400 point x1 • IR Obstacle Sensor module x2 • IR Line-follower sensor module x2 • HC-SR04 Ultrasonic Sensor x1 • Servo SG90 x1 • Bluetooth Module HC-06 x1 • Jumper wires M-M and M-F (30 pcs each) • 4xAA Battery Holder x1 • LED assortment + 220Ω x5 • Push-button x2	1. Obstacle-Avoidance Robot Ultrasonic + servo scanner auto-navigates 2. Line-Follower Robot IR sensors follow black tape on surface 3. Bluetooth-Controlled Car Drive robot from phone via HC-06 app 4. Object-Sorting Robot Colour sensor + servo arm sorts blocks 5. Maze-Solving Robot IR + ultrasonic algorithm navigation 6. Wall-Following Robot Maintain fixed distance from wall 7. Gesture-Controlled Robot MPU-6050 glove controls direction 8. Voice-Controlled Robot Bluetooth + voice commands from phone

Kit 4 — Sensor & Data Logging Kit <i>Measure everything · Ages 16+ · Final year / research projects</i>	
Components Included	Projects You Can Build (8)

• Arduino Mega 2560 x1 • DHT22 Temp and Humidity x1 • BMP280 Pressure and Altitude x1 • DS18B20 Waterproof Temp Probe x1 • MQ-135 Air Quality Sensor x1 • Rain Sensor Module x1 • Current Sensor ACS712 20A x1 • Load Cell 5kg + HX711 module x1 • MPU-6050 Gyro + Accelerometer x1 • Pulse Oximeter MAX30102 x1 • 0.96 inch OLED display x1 • MicroSD card module x1 • MicroSD card 8GB x1 • DS3231 Real-Time Clock module x1 • Jumper wires full set • Breadboard 830 point x1 • 9V power adapter x1	1. Multi-Parameter Weather Station Logs temp, humidity, pressure, rain to SD 2. Smart Energy Meter ACS712 measures current, logs consumption 3. Digital Smart Weighing Scale Load cell + HX711 displays weight on OLED 4. Health Monitoring Band MAX30102 SpO2 + heart rate logged with RTC 5. Earthquake Pre-Alert System MPU-6050 detects vibration spikes with timestamp 6. Air Quality Data Logger MQ-135 + BMP280 log air data every minute 7. Cold Chain Monitor DS18B20 logs food/medicine storage temperature 8. Posture Correction Device MPU-6050 alerts when back angle is wrong
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Kit 5 — Wireless Communication & GPS Kit <i>Connect and locate · Ages 17+ · Advanced / Hackathon projects</i>	
Components Included	Projects You Can Build (8)
• ESP32 Dev Board x1 • SIM800L GSM Module x1 • NEO-6M GPS Module x1 • HC-05 Bluetooth Module x1 • NRF24L01+PA+LNA RF Module x2 • RC522 RFID Module x1 • RFID cards + key fobs x5 • 0.96 inch OLED display x1 • Active Buzzer x1 • Relay module 2-channel x1 • Breadboard 830 point x1 • Jumper wires full set 100 pcs • 18650 cell + holder x1 • TP4056 USB-C charging module x1 • SIM card (user-provided)	1. GPS Vehicle Tracker NEO-6M + SIM800L sends location via SMS 2. RFID Smart Attendance RC522 logs attendance to cloud via ESP32 3. Wireless Multi-Room Sensor Network Two NRF24L01 nodes share sensor data 4. GSM Security Alarm PIR + SIM800L calls/SMS on intrusion 5. Bluetooth Walkie-Talkie Two HC-05 boards relay messages 6. Smart Access Control RFID lock + Bluetooth monitor + cloud log 7. Pet and Luggage Tracker GPS + GSM periodic SMS location tracker 8. Emergency SOS Device Button + GPS + SIM800L sends coordinates

Kit 6 — Display & User Interface Kit <i>Show it off · Ages 14+ · Exhibition / hackathon demos</i>	
Components Included	Projects You Can Build (8)

• Arduino Uno R3 x1 • 16x2 LCD I2C x1 • 0.96 inch OLED I2C x1 • 2.4 inch TFT SPI colour display x1 • TM1637 4-digit 7-segment module x1 • 8x8 Dot Matrix LED MAX7219 x1 • WS2812B Addressable RGB LED strip 8 LEDs x1 • Potentiometer 10kΩ x2 • Push-button x4 • Passive buzzer x1 • DHT11 sensor x1 • Breadboard 830 point x1 • Jumper wires full set • USB cable x1	1. Digital Alarm Clock DS3231 + TM1637 displays time, buzzer alarm 2. Mini Weather Display DHT11 + colour TFT shows data graphically 3. Scrolling LED Matrix Sign Custom messages scroll on 8x8 matrix 4. RGB Mood Lamp WS2812B cycles colours, pot controls speed 5. Simple Game Console 2-button game on OLED display 6. Animated Countdown Timer Colour TFT countdown with progress bar 7. Music Visualiser Sound sensor drives WS2812B intensity 8. Multi-sensor Dashboard TFT shows live readings from sensors
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Kit 7 — Power Electronics & Circuit Design Kit <i>Understand power · Ages 16+ · Electronics and EEE students</i>	
Components Included	Projects You Can Build (8)
• LM2596 Adjustable Buck Converter x1 • MT3608 Boost Converter Module x1 • L7805 Voltage Regulator TO-220 x2 • AMS1117 3.3V Regulator x3 • 555 Timer IC NE555 x3 • LM358 Op-Amp x3 • LM324 Quad Op-Amp x2 • BC547 NPN Transistor x10 • TIP122 Darlington NPN x3 • IRFZ44N N-MOSFET x3 • 1N4007 Rectifier Diode x10 • 1N5819 Schottky Diode x5 • Zener Diodes 3.3V, 5V, 12V x3 each • Resistor assortment 220Ω-100kΩ (200 pcs) • Capacitor assortment 100nF-470uF (50 pcs) • Vero Stripboard 10x15cm x2 • Digital Multimeter x1 • Breadboard 830 point x1 • Jumper wires full set	1. Variable Power Supply LM2596 + voltmeter display, 1.25-35V adjustable 2. PWM Motor Speed Controller 555 timer PWM + MOSFET drives DC motor 3. Audio Amplifier Circuit LM358 amplifies mic signal to 8Ω speaker 4. LED Dimmer Circuit 555 PWM + MOSFET dims LED strip smoothly 5. Transistor-based Switch BC547/TIP122 switch relay from 3.3V logic 6. Voltage Regulator Board L7805 + filter caps make clean 5V from 12V 7. Comparator Circuit LM358 compares voltages, triggers relay/LED 8. Solar Charge Controller MOSFET + Zener + op-amp basic MPPT circuit

Kit 8 — Mini Drone Build Kit <i>Take flight · Ages 18+ · Final year / aerospace projects</i>	
Components Included	Projects You Can Build (5)

- F450 Quadcopter Frame Kit 450mm x1
- Brushless Motor 2212 920KV x4
- 30A BLHeli ESC x4
- Propeller 1045 CW+CCW pairs x2 sets
- Flysky FS-i6 6-CH Transmitter + FS-IA6 Receiver x1
- LiPo Battery 7.4V 2200mAh 2S 25C XT60 x1
- iMax B3 Pro LiPo Balance Charger x1
- Flight Controller F3/F4 compatible x1
- XT60 connector pair x2
- Power Distribution Board PDB x1
- Zip ties + standoffs hardware set
- LiPo battery voltage alarm buzzer x1

1. Build and Fly Quadcopter

Full F450 assembly — frame, motors, ESC, FC, remote

2. Altitude Hold Drone

Barometric sensor + FC firmware for stable hover

3. FPV Camera Drone

ESP32-CAM + video transmitter for first-person view

4. GPS Return-to-Home Drone

NEO-M8N GPS + ArduPilot autonomous return

5. Payload Drop Drone

Servo mechanism drops payload on command